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GROUP ASSIGNMENT

MULTIVARIABLE CALCULUS AND DIFFERENTIAL EQUATIONS

(MATH 8214)

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GROUP B -TEAM 1

ACADEMIC YEAR: 2024-2025

SEMESTER: I

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QUESTION 1:

Consider the triangle with vertices:

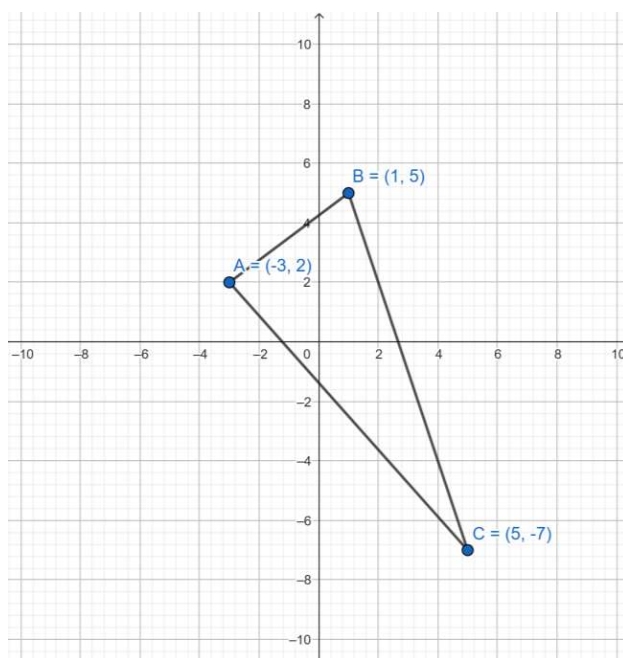
A(-3,2), B(1,5), C(-5,7) (T1)

- Write down all 8 equations of the side AB
- Find the perimeter, area, angles, centroid, orthocenter, incenter, and circumcenter of the triangle
- Write down the equation of median, altitude and angle bisector line dropped from vertex A
- Evaluate the length of the median, altitude, angle bisector line dropped from the vertex A
- Write down the parametric and analytic equations of the circumcircle and incircle
- Evaluate the perimeter and area of the circumcircle and in circle
- Use 2D-Geogebra and plot the triangle with its incircle and circumcircle

Answer:

Consider the triangle with vertices

A (-3,2), B (1,5), C (5, -7)



Vectors:

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{bmatrix} 1 - (-3) \\ 5 - 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$|\overrightarrow{AB}| = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{bmatrix} 5 - (-3) \\ -7 - 2 \end{bmatrix} = \begin{bmatrix} 8 \\ -9 \end{bmatrix}$$

$$|\overrightarrow{AC}| = \sqrt{8^2 + (-9)^2} = \sqrt{64 + 81} = \sqrt{145}$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{bmatrix} 5 - 1 \\ -7 - 5 \end{bmatrix} = \begin{bmatrix} 4 \\ -12 \end{bmatrix}$$

$$|\overrightarrow{BC}| = \sqrt{4^2 + (-12)^2} = \sqrt{16 + 144} = \sqrt{160}$$

A. Write down all 8 equations of the side AB

i. Vector equation:

$$\overrightarrow{AB}: \vec{r} = \overrightarrow{OA} + \lambda \overrightarrow{AB}$$

ii. Parametric equation:

$$AB: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

iii. Canonical equation:

$$AB: \frac{x+3}{4} = \frac{y-2}{3}$$

iv. Analytical implicit equation:

$$AB : 3x + 9 - 4y + 8 = 0$$

$$AB : 3x - 4y + 17 = 0$$

v. Analytical explicit equation:

$$AB: y = \frac{3}{4}x + \frac{17}{4}$$

vi. Normal equation:

$$x \cos \alpha + y \sin \alpha - p = 0$$

$$\frac{A}{\sqrt{A^2+B^2}} = \cos \alpha$$

$$\frac{B}{\sqrt{A^2+B^2}} = \sin \alpha$$

Where A and B are 3 and -4 respectively

$$\cos \alpha = \frac{3}{\sqrt{3^2+4^2}} = \frac{3}{5}$$

$$\sin \alpha = \frac{-4}{\sqrt{3^2+4^2}} = \frac{-4}{5}$$

$$p = \frac{C}{\sqrt{A^2+B^2}} = \frac{17}{\sqrt{3^2+4^2}} = \frac{17}{5}$$

$$p = \frac{17}{5}$$

So, the normal equation is

$$AB: \frac{3}{5}x - \frac{4}{5}y - \frac{17}{5} = 0$$

vii. Polar equation:

$$r = \frac{p}{\cos(\alpha - \theta)}$$

$$r = \frac{17}{5\cos(\alpha - \theta)}$$

viii. Intercept formula :

$$AB: \frac{x}{a} + \frac{y}{b} = 1$$

$$\text{X-intercept: } \begin{cases} y = \frac{3}{4}x + \frac{17}{4} \\ y = 0 \end{cases} \quad p\left(\frac{-17}{3}, 0\right)$$

$$\text{Y-intercept: } \begin{cases} y = \frac{3}{4}x + \frac{17}{4} \\ x = 0 \end{cases} \quad p\left(0, \frac{17}{4}\right)$$

$$AB: \frac{x}{\frac{-17}{3}} + \frac{y}{\frac{17}{4}} = 1$$

$$AB: -\frac{3x}{17} + \frac{4y}{17} = 1$$

B. Find the perimeter, area, angles, centroid, orthocenter, incenter and circumcenter of the triangle

i) Perimeter

$$P_{ABC} = (\|\overrightarrow{AB}\| + \|\overrightarrow{BC}\| + \|\overrightarrow{AC}\|) \text{ LU}$$

$$P_{ABC} = (5 + 4\sqrt{10} + \sqrt{145}) \text{ LU}$$

ii) Area

$$S_{ABC} = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix} \text{ Squ} = \frac{1}{2} \begin{vmatrix} -3 & 1 & 5 & -3 \\ 2 & 5 & -7 & 2 \end{vmatrix} \text{ Squ}$$

$$S_{ABC} = \frac{1}{2} |-12 - 48| = 30 \text{ Squ}$$

iii) Angles

Angle $\hat{A}$

$$\cos \hat{A} = \frac{\overrightarrow{AB} \cdot \overrightarrow{AC}}{\|\overrightarrow{AB}\| \|\overrightarrow{AC}\|} = \frac{(4 \times 8) + (3 \times (-9))}{5\sqrt{145}} = \frac{1}{\sqrt{145}} = \frac{\sqrt{145}}{145}$$

$$\hat{A} = \cos^{-1}\left(\frac{\sqrt{145}}{145}\right)$$

Angle $\hat{B}$

$$\cos \hat{B} = \frac{\vec{BA} \cdot \vec{BC}}{\|\vec{BA}\| \|\vec{BC}\|} = \left(\frac{(4 \times (-4)) + ((-1) \times (-3))}{5(4\sqrt{10})} \right) = \frac{20}{20\sqrt{10}} = \frac{\sqrt{10}}{10}$$

$$\hat{B} = \cos^{-1} \left(\frac{\sqrt{10}}{10} \right)$$

Angle $\hat{C}$

$$\cos \hat{C} = \frac{\vec{CB} \cdot \vec{CA}}{\|\vec{CB}\| \|\vec{CA}\|} = \frac{((4 \times (-8)) + (12 \times 9))}{(\sqrt{145}) \times (4\sqrt{10})} = \frac{140}{20\sqrt{58}} = 7 \frac{\sqrt{58}}{58}$$

$$\hat{C} = \cos^{-1} \left(7 \frac{\sqrt{58}}{58} \right)$$

iv) Centroid

$$G \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right) = G \left(\frac{-3 + 1 + 5}{3}, \frac{2 + 5 - 7}{3} \right) = G \left(\frac{3}{3}, \frac{0}{3} \right)$$

G (1, 0)

v) Incenter

$$I \left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c} \right)$$

$$I \left(\frac{-3\sqrt{160} + \sqrt{145} + 25}{\sqrt{160} + \sqrt{145} + 5}, \frac{2\sqrt{160} + 5\sqrt{145} - 35}{\sqrt{160} + \sqrt{145} + 5} \right)$$

vi) Orthocenter

$$H \left(\frac{\begin{vmatrix} y_1^2 + x_2x_3 & y_1 & 1 \\ y_2^2 + x_1x_3 & y_2 & 1 \\ y_3^2 + x_1x_2 & y_3 & 1 \end{vmatrix}}{2S_{ABC}}, \frac{\begin{vmatrix} x_1 & x_1^2 + y_2y_3 & 1 \\ x_2 & x_2^2 + y_1y_3 & 1 \\ x_3 & x_3^2 + y_1y_2 & 1 \end{vmatrix}}{2S_{ABC}} \right)$$

$$H \left(\frac{\begin{vmatrix} 4+5 & 2 & 1 \\ 25-15 & 5 & 1 \\ 49-3 & -7 & 1 \end{vmatrix}}{60}, \frac{\begin{vmatrix} -3 & 9-35 & 1 \\ 1 & 1-14 & 1 \\ 5 & 25+10 & 1 \end{vmatrix}}{60} \right) = H \left(\frac{\begin{vmatrix} 9 & 2 & 1 \\ 10 & 5 & 1 \\ 46 & -7 & 1 \end{vmatrix}}{60}, \frac{\begin{vmatrix} -3 & -26 & 1 \\ 1 & -13 & 1 \\ 5 & 35 & 1 \end{vmatrix}}{60} \right)$$

$$H \left(-2, \frac{7}{3} \right)$$

vii) Circumcenter

$$O \left(\frac{\begin{vmatrix} y_1 & x_1^2 + y_1^2 & 1 \\ y_2 & x_2^2 + y_2^2 & 1 \\ y_3 & x_3^2 + y_3^2 & 1 \end{vmatrix}}{4S_{ABC}}, \frac{\begin{vmatrix} x_1^2 + y_1^2 & x_1 & 1 \\ x_2^2 + y_2^2 & x_2 & 1 \\ x_3^2 + y_3^2 & x_3 & 1 \end{vmatrix}}{4S_{ABC}} \right)$$

$$\mathbf{O} \left(\frac{\begin{vmatrix} 13 & 2 & 1 \\ 26 & 5 & 1 \\ 74 & -7 & 1 \end{vmatrix}}{120}, \frac{\begin{vmatrix} 13 & -3 & 1 \\ 26 & 1 & 1 \\ 74 & 5 & 1 \end{vmatrix}}{120} \right) = \mathbf{O} \left(\frac{300}{120}, -\frac{140}{120} \right)$$

$$\mathbf{O} \left(\frac{5}{2}, -\frac{7}{6} \right)$$

Euler's formula:

$$\overrightarrow{HG} = 2\overrightarrow{GO}$$

$$\overrightarrow{HG} = \overrightarrow{OG} - \overrightarrow{OH} = \begin{bmatrix} 1 - (-2) \\ 0 - \frac{7}{3} \end{bmatrix} = \begin{bmatrix} 3 \\ -\frac{7}{3} \end{bmatrix}$$

$$2\overrightarrow{GO} = 2(\overrightarrow{OO} - \overrightarrow{OG}) = 2 \begin{bmatrix} \frac{5}{2} - 1 \\ -\frac{7}{6} - 0 \end{bmatrix} = 2 \begin{bmatrix} \frac{3}{2} \\ -\frac{7}{6} \end{bmatrix} = \begin{bmatrix} 3 \\ -\frac{7}{3} \end{bmatrix}$$

C. Write down the equation of the median, altitude, angle bisector line dropped from A

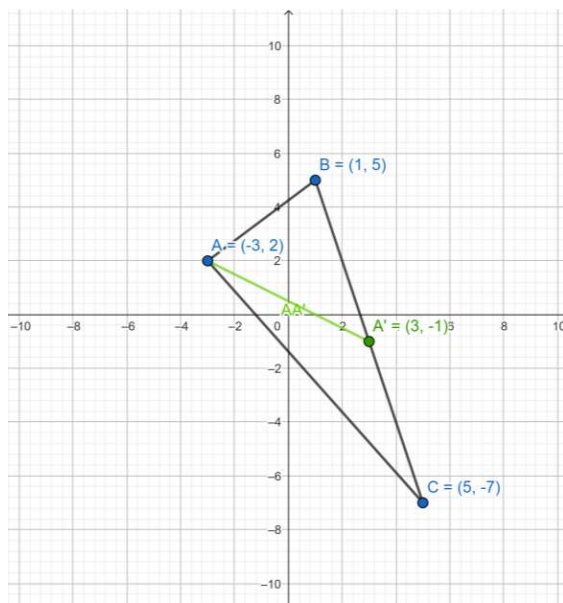
i. Equation of the median dropped from A

A (-3,2)

Let A' be the median dropped from vertex A

$$A' \left(\frac{x_C + x_B}{2}, \frac{y_C + y_B}{2} \right) = A' \left(\frac{5+1}{2}, \frac{-7+5}{2} \right) = A' (3, -1)$$

$$\overrightarrow{AA'} = \overrightarrow{OA'} - \overrightarrow{OA} = \begin{bmatrix} 3 - (-3) \\ -1 - 2 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$



It's vector equation

$$m_a: \vec{v} = \overrightarrow{OA} + \lambda \overrightarrow{AA'}$$

Parametric equation

$$m_a: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$

Canonical equation

$$m_a: \frac{x+3}{6} = \frac{y-2}{-3}$$

Analytical implicit equation:

$$m_a: -3x - 9 - 6y + 12 = 0$$

$$m_a: 3x + 6y - 3 = 0$$

The length of the median from A

$$m_a = \frac{1}{2} \sqrt{2(b^2 + c^2) - a^2} \text{ LU}$$

$$m_a = \frac{1}{2} \sqrt{2(145 + 25) - 160} \text{ LU}$$

$$m_a = \frac{1}{2} \sqrt{180} \text{ LU} = 3\sqrt{5} \text{ LU}$$

ii. Equation of altitude dropped from A

$$h_a \perp BC : 12x + 4y - 32 = 0$$

$$h_a \perp BC : y = -3x + 8$$

$$h_a: y = \frac{1}{3}x + m' \text{ since } A(-3,2)$$

$$m' = 3$$

$$h_a: y = \frac{1}{3}x + 3$$

$$h_a: 3y - x - 9 = 0$$

iii. Length of altitude from A

$$h_a = \frac{2S_{ABC}}{a} \text{ LU}$$

$$h_a = \frac{2 \times 30}{4\sqrt{10}} \text{ LU} = \frac{3\sqrt{10}}{2} \text{ LU}$$

iv. Equation of angle bisector line from A (Ia)

Equation of AB and AC are $3x - 4y + 17 = 0$ and $9x + 8y + 11 = 0$ respectively

$$l_{i,e}: \frac{A_1x+B_1y+C_1}{\sqrt{A_1^2+B_1^2}} = \pm \frac{A_2x+B_2y+C_2}{\sqrt{A_2^2+B_2^2}}$$

$$l_{i,e}: \frac{3x-4y+17}{\sqrt{25}} = \pm \frac{9x+8y+11}{\sqrt{145}}$$

$$l_1: \frac{3x-4y+17}{5} = \frac{9x+8y+11}{\sqrt{145}}$$

$$l_1: 3\sqrt{145}x - 4\sqrt{145}y + 17\sqrt{145} = 45x + 40y + 55$$

$$l_1: -4\sqrt{145}y - 40y = -3\sqrt{145}x + 45x + 55 + 17\sqrt{145}$$

$$l_1: y(-4\sqrt{145} - 40) = (-3\sqrt{145} + 45)x + 55 + 17\sqrt{145}$$

$$l_1: y = \left(\frac{-3\sqrt{145}+45}{-4\sqrt{145}-40}\right)x + \left(\frac{55+17\sqrt{145}}{-4\sqrt{145}-40}\right) \text{ rejected}$$

$$l_2: \frac{3x-4y+17}{5} = -\left(\frac{9x+8y+11}{\sqrt{145}}\right)$$

$$l_2: 3\sqrt{145}x - 4\sqrt{145}y + 17\sqrt{145} = -45x - 40y - 55$$

$$l_2: y(-4\sqrt{145} + 40) = -3\sqrt{145} - 45)x - 55 - 17\sqrt{145}$$

$$l_2: y = \left(\frac{-3\sqrt{145}-45}{-4\sqrt{145}+40}\right)x + \left(\frac{-55-17\sqrt{145}}{-4\sqrt{145}+40}\right)$$

Since the gradient of line 2 is positive thus we take the equation of line 2.

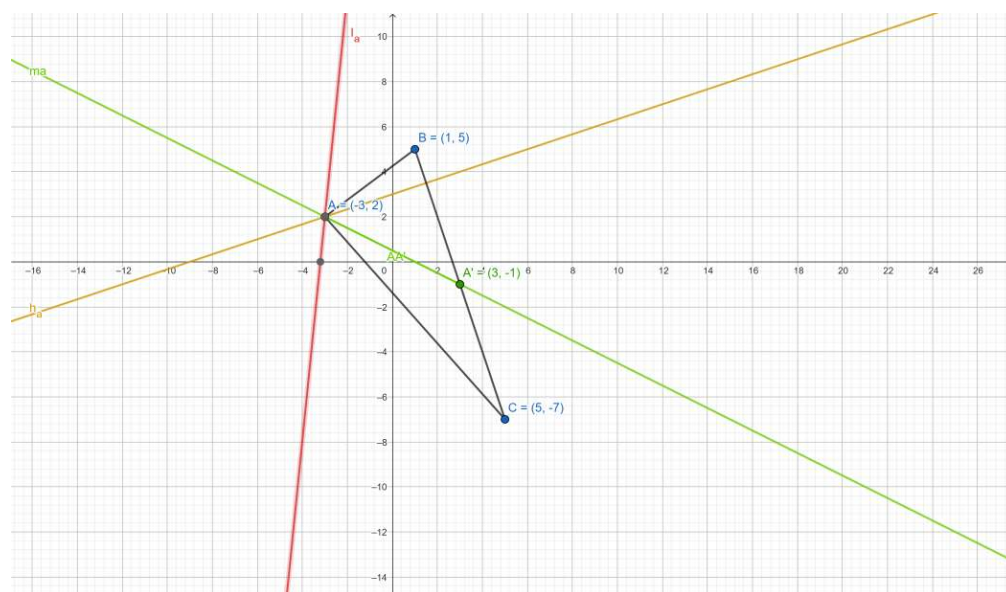
$$l_a: (3\sqrt{145} + 45)x + (-4\sqrt{145} + 40)y + 17\sqrt{145} + 55 = 0 \text{ that is the equation of } l_a$$

v. The length of l_a

$$l_a = \frac{1}{b+c} \sqrt{bc[(b+c)^2 - a^2]} LU$$

$$l_a = \frac{1}{5 + \sqrt{145}} \sqrt{5\sqrt{145}[(5 + \sqrt{145})^2 - (\sqrt{160})^2]} LU$$

Graph showing the median, altitude and angel bisector line dropped from vertex A.



D. Write down the parametric and analytical equation of the circumcircle and incircle

Parametric Equation of incircle:

$$\begin{cases} x = I_x + r \cos t \\ y = I_y + r \sin t \end{cases}$$

$$r = \frac{2S_{ABC}}{P_{ABC}} LU \text{ where } r \text{ is the inradius}$$

$$r = \frac{60}{5+4\sqrt{10}+\sqrt{145}} LU$$

$$\Gamma: \begin{cases} x = \frac{-3\sqrt{160}+\sqrt{145}+25}{\sqrt{160}+\sqrt{145}+5} + 2 \cos t \\ y = \frac{2\sqrt{160}+5\sqrt{145}-35}{\sqrt{160}+\sqrt{145}+5} + 2 \sin t \end{cases}$$

Analytical Equation of incircle:

$$\Gamma: (x - I_x)^2 + (y - I_y)^2 = R^2$$

$$\Gamma: \left(x - \frac{-3\sqrt{160}+\sqrt{145}+25}{\sqrt{160}+\sqrt{145}+5}\right)^2 + \left(y - \frac{2\sqrt{160}+5\sqrt{145}-35}{\sqrt{160}+\sqrt{145}+5}\right)^2 = \left(\frac{60}{5+4\sqrt{10}+\sqrt{145}}\right)^2$$

Parametric Equation of circumcircle:

$$\Gamma: \begin{cases} x = O_x + R \cos t \\ y = O_y + R \sin t \end{cases}$$

$$R = \frac{abc}{4S_{ABC}} \text{ where } R \text{ is the circumradius}$$

$$R = \frac{5 \times 4\sqrt{10} \times \sqrt{145}}{120} = \frac{20\sqrt{1450}}{120} = \frac{5\sqrt{58}}{6} LU$$

$$\begin{cases} x = \frac{-5}{2} + \frac{5\sqrt{58}}{6} \cos t \\ y = \frac{7}{6} + \frac{5\sqrt{58}}{6} \sin t \end{cases} \text{ where } t \text{ is the parameter, and it varies from } 0 \text{ to } 2\pi$$

Analytical Equation of circumcircle:

$$\Gamma: (x - O_x)^2 + (y - O_y)^2 = R^2$$

$$\Gamma: \left(x - \frac{5}{2}\right)^2 + \left(y + \frac{7}{6}\right)^2 = \left(\frac{5\sqrt{58}}{6}\right)^2$$

E. Evaluate the perimeter and the Area of circumcircle and incircle:

The perimeter of incircle

$$P_\gamma = 2r\pi LU$$

$$P_{\gamma} = 2 \times \frac{60}{5 + 4\sqrt{10} + \sqrt{145}} \pi LU = \frac{120\pi}{5 + 4\sqrt{10} + \sqrt{145}} LU$$

The Area of incircle

$$A_{\gamma} = \pi r^2 \text{ square units}$$

$$A_{\gamma} = \pi \left(\frac{60}{5 + 4\sqrt{10} + \sqrt{145}} \right)^2 \text{ squ}$$

The perimeter of circumcircle

$$P_{\Gamma} = 2R\pi LU$$

$$P_{\Gamma} = 2 \times \frac{5\sqrt{58}}{6} \pi LU$$

$$P_{\Gamma} = \frac{5\pi \sqrt{58}}{3} LU$$

The Area of circumcircle

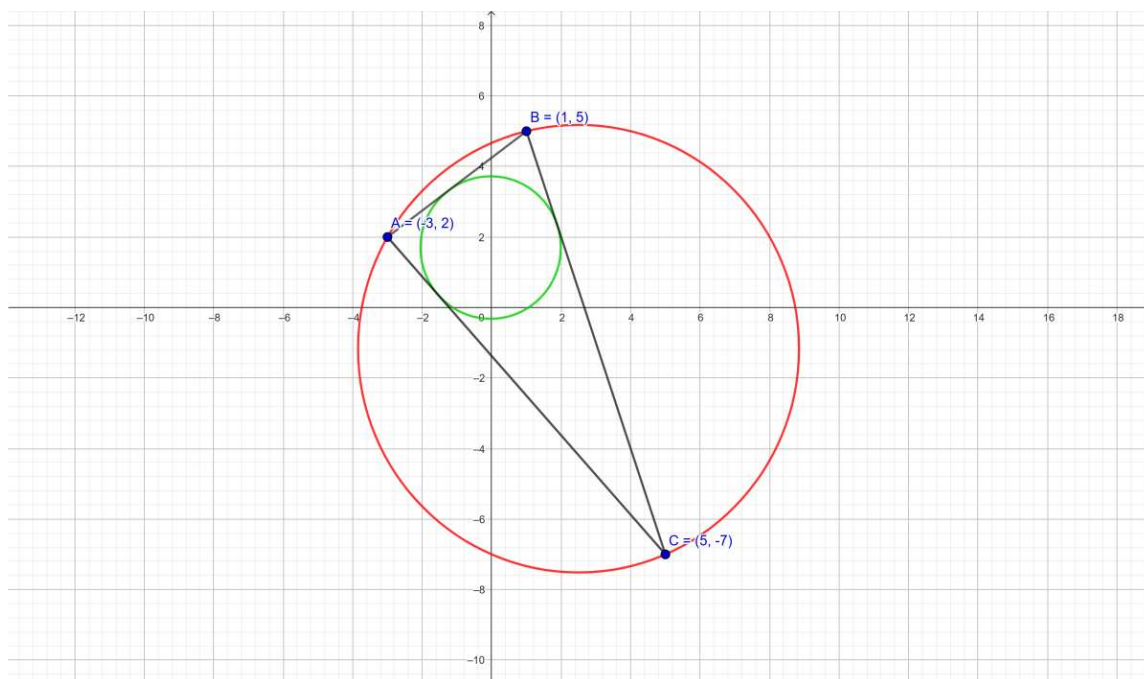
$$A_{\Gamma} = \pi R^2 \text{ Squ}$$

$$A_{\Gamma} = \pi \left(\frac{5\sqrt{58}}{6} \right)^2 \text{ Squ}$$

F. Use 2-D GeoGebra and plot the triangle with its incircle and circumcircle

$$\Gamma: \left(x - \frac{-3\sqrt{160} + \sqrt{145} + 2}{\sqrt{160} + \sqrt{145} + 5} \right)^2 + \left(y - \frac{2\sqrt{160} + 5\sqrt{145} - 35}{\sqrt{160} + \sqrt{145} + 5} \right)^2 = \left(\frac{60}{5 + 4\sqrt{10} + \sqrt{145}} \right)^2 \text{ Equation of incircle}$$

$$\Gamma: \left(x - \frac{5}{2} \right)^2 + \left(y + \frac{7}{6} \right)^2 = \left(\frac{5\sqrt{58}}{6} \right)^2 \text{ Equation of circumcircle}$$



QUESTION 2:

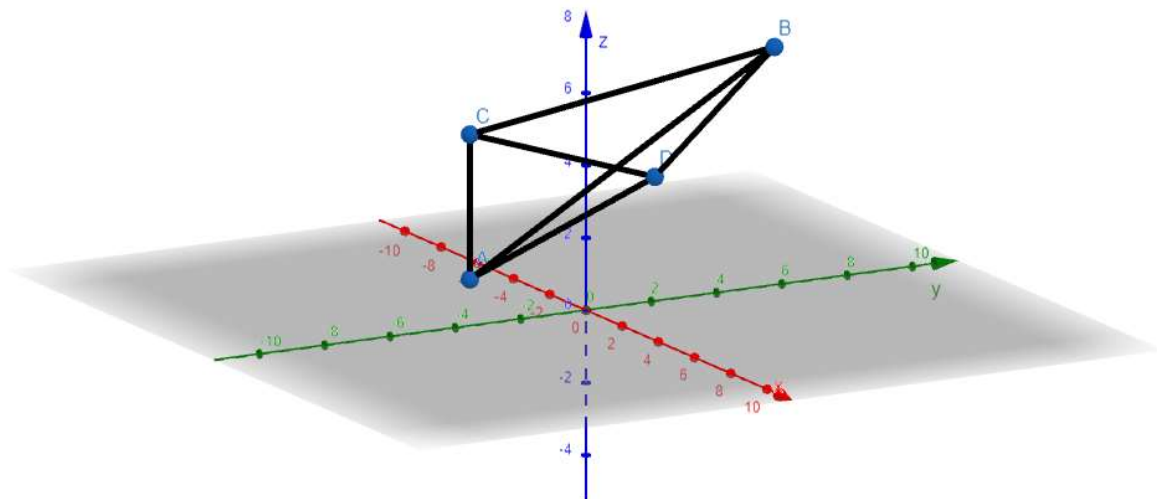
Consider the tetrahedron with vertices:

$A(-1, -3, 1), B(5, 3, 8), C(-1, -3, 5), D(2, 1, 4)$ (T1)

- Evaluate the area and perimeter of the face ABC
- Calculate the length of the median and altitude dropped from the vertex A of the face ABC
- Evaluate the total surface area and the volume of the tetrahedron
- Write the equation and calculate the length of altitude of the tetrahedron from the vertex A
- Determine the angle between the edge AD and face ABC
- Find the distance between edges AD and BC
- Write down the equations and evaluate the area and volume of the circumsphere
- Use 3D Geo-Gebra and plot the tetrahedron and its circumsphere

Answer:

Consider the tetrahedron with vertices $A(-1, -3, 1)$, $B(5, 3, 8)$, $C(-1, -3, 5)$ and $D(2, 1, 4)$



Vectors of the face ABC:

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{bmatrix} 5 - (-1) \\ 3 - (-3) \\ 8 - 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ 7 \end{bmatrix}$$

$$|\overrightarrow{AB}| = \sqrt{6^2 + 6^2 + 7^2} = \sqrt{36 + 36 + 49} = \sqrt{121} = 11$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{bmatrix} -1 - (-1) \\ -3 - (-3) \\ 5 - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$$

$$|\overrightarrow{AC}| = \sqrt{0^2 + 0^2 + 4^2} = \sqrt{16} = 4$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{bmatrix} -1 - 5 \\ -3 - 3 \\ 5 - 8 \end{bmatrix} = \begin{bmatrix} -6 \\ -6 \\ -3 \end{bmatrix}$$

$$|\overrightarrow{BC}| = \sqrt{(-6)^2 + (-6)^2 + (-3)^2} = \sqrt{81} = 9$$

A. Evaluate the area and perimeter of the face ABC

i) **Perimeter:**

$$P_{ABC} = (\|\overrightarrow{AB}\| + \|\overrightarrow{AC}\| + \|\overrightarrow{BC}\|) \text{ LU}$$

$$P_{ABC} = (11 + 4 + 9) \text{ LU}$$

$$P_{ABC} = 24 \text{ LU}$$

ii) **Area**

$$S_{ABC} = \frac{1}{2} \|\overrightarrow{AB}, \overrightarrow{AC}\| \text{ Squ}$$

$$S_{ABC} = \frac{1}{2} \left\| \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & 6 & 7 \\ 0 & 0 & 4 \end{bmatrix} \right\| \text{ Squ}$$

$$S_{ABC} = \frac{1}{2} \left(\left\| \begin{bmatrix} 6 & 7 \\ 0 & 4 \end{bmatrix} \vec{i} - \begin{bmatrix} 6 & 7 \\ 0 & 4 \end{bmatrix} \vec{j} + \begin{bmatrix} 6 & 6 \\ 0 & 0 \end{bmatrix} \vec{k} \right\| \right) \text{ Squ}$$

$$S_{ABC} = \frac{1}{2} \| 24\vec{i} - 24\vec{j} \| \text{ Squ}$$

$$S_{ABC} = \frac{1}{2} \sqrt{24^2 + 24^2} \text{ Squ}$$

$$\text{Area of face ABC, } S_{ABC} = 12\sqrt{2} \text{ Squ}$$

B. Calculate the length of the median and altitude dropped from vertex A of the face ABC

Let $\mathbf{m}_a$ be the median dropped from vertex A of the face ABC and A' is the Midpoint of B and C

i) **Length of median:**

$$A' \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

$$A' \left(\frac{5-1}{2}, \frac{3-3}{2}, \frac{8+5}{2} \right)$$

$$A' \left(2, 0, \frac{13}{2} \right)$$

$$\overrightarrow{AA'} = \overrightarrow{OA'} - \overrightarrow{OA} = \begin{bmatrix} 2+1 \\ 0+3 \\ \frac{13}{2}-1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ \frac{11}{2} \end{bmatrix}$$

$$\mathbf{m}_a: \|\overrightarrow{AA'}\| = \sqrt{3^2 + 3^2 + \left(\frac{11}{2}\right)^2} = \frac{\sqrt{193}}{2} \text{ L.U}$$

ii) **Length of altitude:**

$$h_a = \frac{2S_{ABC}}{a} = \frac{2 \times 12\sqrt{2}}{9} = \frac{8\sqrt{2}}{3} \text{ LU}$$

C. Evaluate the total surface area and volume of the tetrahedron

i) **Total surface area:**

$$TSA = (S_{ABC} + S_{ABD} + S_{ACD} + S_{BCD}) \text{ Squ}$$

$$S_{ABC} = 12\sqrt{2} \text{ Squ}$$

$$S_{ABD} = \frac{1}{2} \| [\overrightarrow{AB}, \overrightarrow{AD}] \| \text{ Squ}$$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{bmatrix} 2+1 \\ 1+3 \\ 4-1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}$$

$$S_{ABD} = \frac{1}{2} \left\| \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & 6 & 7 \\ 3 & 4 & 3 \end{bmatrix} \right\| \text{ Squ}$$

$$= \frac{1}{2} \left(\left\| \begin{bmatrix} 6 & 7 \\ 4 & 3 \end{bmatrix} \vec{i} - \begin{bmatrix} 6 & 7 \\ 3 & 3 \end{bmatrix} \vec{j} + \begin{bmatrix} 6 & 6 \\ 3 & 4 \end{bmatrix} \vec{k} \right\| \right) \text{ Squ}$$

$$= \frac{1}{2} \| -10\vec{i} + 3\vec{j} + 6\vec{k} \| \text{ Squ}$$

$$= \frac{1}{2} \sqrt{(-10)^2 + 3^2 + 6^2} \text{ Squ}$$

$$\text{Area of face ABD, } S_{ABD} = \frac{1}{2} \sqrt{145} \text{ Squ}$$

$$\text{Area of ACD} = S_{ACD} = \frac{1}{2} \| [\vec{AC}, \vec{AD}] \| \text{Squ}$$

$$S_{ACD} = \frac{1}{2} \left\| \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & 4 \\ 3 & 4 & 3 \end{vmatrix} \right\| \text{Squ}$$

$$= \frac{1}{2} \left(\left\| \begin{vmatrix} 0 & 4 \\ 4 & 3 \end{vmatrix} \vec{i} - \begin{vmatrix} 0 & 4 \\ 3 & 3 \end{vmatrix} \vec{j} + \begin{vmatrix} 0 & 0 \\ 3 & 4 \end{vmatrix} \vec{k} \right\| \text{Squ} \right.$$

$$= \frac{1}{2} \| -16\vec{i} + 12\vec{j} + 0\vec{k} \| \text{Squ}$$

$$= \frac{1}{2} \sqrt{(-16)^2 + 12^2 + 0^2} \text{Squ}$$

$$S_{ACD} = \frac{1}{2} \sqrt{400} \text{ Squ}$$

$$\text{Area of face ACD} , S_{ACD} = 10 \text{ Squ}$$

$$S_{BCD} = \frac{1}{2} \| [\vec{BC}, \vec{BD}] \| \text{Squ}$$

$$\vec{BD} = \vec{OD} - \vec{OB} = \begin{bmatrix} 2-5 \\ 1-3 \\ 4-8 \end{bmatrix} = \begin{bmatrix} -3 \\ -2 \\ -4 \end{bmatrix}$$

$$S_{BCD} = \frac{1}{2} \left\| \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -6 & -6 & -3 \\ -3 & -2 & -4 \end{vmatrix} \right\| \text{Squ}$$

$$= \frac{1}{2} \left(\left\| \begin{vmatrix} -6 & -3 \\ -2 & -4 \end{vmatrix} \vec{i} - \begin{vmatrix} -6 & -3 \\ -3 & -4 \end{vmatrix} \vec{j} + \begin{vmatrix} -6 & -6 \\ -3 & -2 \end{vmatrix} \vec{k} \right\| \text{Squ} \right.$$

$$= \frac{1}{2} \| 18\vec{i} - 15\vec{j} - 6\vec{k} \| \text{Squ}$$

$$= \frac{1}{2} \sqrt{18^2 + (-15)^2 + (-6)^2} \text{Squ}$$

$$\text{Area of face BCD} , S_{BCD} = \frac{1}{2} \sqrt{585} \text{ squ}$$

$$TSA = (S_{ABC} + S_{ABD} + S_{ACD} + S_{BCD}) \text{ squ}$$

$$TSA = \left(\frac{1}{2} \sqrt{145} + 12\sqrt{2} + 10 + \frac{1}{2} \sqrt{585} \right) \text{ squ}$$

ii) **Volume:**

$$\text{Volume} = \frac{1}{6} | \vec{AB} \cdot (\vec{AC} \times \vec{AD}) | \text{cu}$$

$$= \frac{1}{6} \begin{vmatrix} 6 & 6 & 7 \\ 0 & 0 & 4 \\ 3 & 4 & 3 \end{vmatrix} \text{cu}$$

$$= \frac{1}{6} |4(24-18)| \text{ cu}$$

Volume, V = 4 cu

D. Equation and length of altitude of the tetrahedron from the vertex A

i) Equation of altitude of the tetrahedron from the vertex A:

$$h_a: \vec{V} = \vec{OA} + \lambda \vec{N}_{ABC}$$

$$\vec{N}_{ABC} = [\vec{AB}, \vec{AC}]$$

$$= \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & 6 & 7 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= 24\vec{i} - 24\vec{j}$$

$$h_a: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 24 \\ -24 \\ 0 \end{bmatrix} \quad \text{parametric equation}$$

$$h_a: \frac{x+1}{24} = \frac{y+3}{-24} = \frac{z-1}{0} \quad \text{canonical equation}$$

ii) Length of altitude of the tetrahedron from the vertex A

equation of plane ABC

$$ABC: \vec{V} = \vec{OA} + \alpha \vec{AB} + \beta \vec{AC}$$

$$ABC: \det(\vec{V} - \vec{OA}, \vec{AB}, \vec{AC}) = 0$$

$$\begin{vmatrix} x+1 & y+3 & z-1 \\ 6 & 6 & 7 \\ 0 & 0 & 4 \end{vmatrix} = 0$$

$$ABC: (x+1) \begin{vmatrix} 6 & 7 \\ 0 & 4 \end{vmatrix} - (y+3) \begin{vmatrix} 6 & 7 \\ 0 & 4 \end{vmatrix} + (z-1) \begin{vmatrix} 6 & 6 \\ 0 & 0 \end{vmatrix} = 0$$

$$24(x+1) - 24(y+3) + 0(z-1) = 0$$

$$ABC: x - y - 2 = 0$$

$$h_a: d(A(-1, -3, 1), \Pi) = \frac{|1(-1) + (-1)(-3) + (0)(1) - 2|}{\sqrt{(1)^2 + (-1)^2 + (0)^2}} = \frac{0}{\sqrt{2}}$$

$$h_a = 0 \text{ LU}$$

(The length of the altitude from A to the face ABC is 0, which means that point A lies on the plane formed by points B and C)

E. Determine the angle between the edge AD and face ABC

Angle:

$$\sin \theta = \frac{\langle \vec{N}_{ABC}, \vec{AD} \rangle}{\|\vec{N}_{ABC}\| \|\vec{AD}\|}$$

$$\sin \theta = \frac{(24 \times 3) + (-24 \times 4) + (0 \times 3)}{(\sqrt{24^2 + (-24)^2}) \times \sqrt{3^2 + 4^2 + 3^2}} = \frac{-24}{48\sqrt{17}}$$

$$\theta = \sin^{-1}\left(\frac{\sqrt{17}}{34}\right)$$

F. Find the distance between edges AD and BC

Distance:

$$d(AD, BC) = \frac{|\langle \vec{AB}, \vec{AD}, \vec{BC} \rangle|}{\|[AD, BC]\|}$$

$$\vec{AD} = \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}$$

$$\vec{BC} = \begin{bmatrix} -1 - 5 \\ -3 - 3 \\ 5 - 8 \end{bmatrix} = \begin{bmatrix} -6 \\ -6 \\ -3 \end{bmatrix}$$

$$\vec{AB} = \begin{bmatrix} 6 \\ 6 \\ 7 \end{bmatrix}$$

$$= \frac{\begin{vmatrix} 6 & 6 & 7 \\ 3 & 4 & 3 \\ -6 & -6 & -3 \end{vmatrix}}{\left\| \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 4 & 3 \\ -6 & -6 & -3 \end{bmatrix} \right\|} = \frac{6 \begin{vmatrix} 4 & 3 \\ -6 & -3 \end{vmatrix} - 6 \begin{vmatrix} 3 & 3 \\ -6 & -3 \end{vmatrix} + 7 \begin{vmatrix} 3 & 4 \\ -6 & -6 \end{vmatrix}}{\sqrt{(6)^2 + (9)^2 + (6)^2}} = \frac{6(6) - 6(9) + 7(6)}{\sqrt{153}}$$

$$d(AD, BC) = \frac{24}{\sqrt{153}} \text{ LU}$$

G. Equation and the area and volume of the circumsphere.

i) Equation:

$$S: x^2 + y^2 + z^2 + 2mx + 2ny + 2pz + q = 0$$

$$\begin{aligned}
S \ni A(-1, -3, 1) : 1 + 9 + 1 - 2m - 6n + 2p + q &= 0 \\
S \ni B(5, 3, 8) : 25 + 9 + 64 + 10m + 6n + 16p + q &= 0 \\
S \ni C(-1, -3, 5) : 1 + 9 + 25 - 2m - 6n + 10p + q &= 0 \\
S \ni D(2, 1, 4) : 4 + 1 + 16 + 4m + 2n + 8p + q &= 0
\end{aligned}$$

$$\begin{cases}
2m + 6n - 2p - q = 1 \dots \dots \dots \text{eqn 1} \\
10m + 6n + 16p + q = -98 \dots \dots \dots \text{eqn 2} \\
2m + 6n - 10p - q = 35 \dots \dots \dots \text{eqn 3} \\
4m + 2n + 8p + q = -21 \dots \dots \dots \text{eqn 4}
\end{cases}$$

From eqn1 and 3

$$2m + 6n - 2p - q = 11$$

—

$$2m + 6n - 10p - q = 35$$

$$8p = -24$$

$$p = -3$$

From eqn2 and 4

$$10m + 6n + 16p + q = -98$$

—

$$4m + 2n + 8p + q = -21$$

$$6m + 4n + 8p = -77 \dots \dots \dots \text{eqn 5}$$

From eqn3 and 4 (add)

$$6m + 8n - 2p = 14 \dots \dots \dots \text{eqn 6}$$

From equation 5 substitute the value of p

$$6m + 4n = -53$$

From equation 6 substitute the value of p

$$6m + 8n = 8$$

From eqn5 and 6 (subtract)

$$-4n = -61$$

$$n = \frac{61}{4} = 15.25$$

substitute the value of n in equation 6

$$m = -19$$

substitute the value of n, m and p in equation 4

$$q = \frac{97}{2}$$

Thus

$$x^2 + y^2 + z^2 + 2(-19)x + 2\left(\frac{61}{4}\right)y + 2(-3)z + \frac{97}{2} = 0$$

$$(x - 19)^2 + \left(y + \frac{61}{4}\right)^2 + (z - 3)^2 = -\frac{97}{2} + 361 + \frac{3721}{16} + 9$$

$$(x - 19)^2 + \left(y + \frac{61}{4}\right)^2 + (z - 3)^2 = \frac{8865}{16}$$

$$(x - 19)^2 + \left(y + \frac{61}{4}\right)^2 + (z - 3)^2 = \left(\frac{\sqrt{8865}}{4}\right)^2 \rightarrow \text{Equation of circumsphere}$$

ii) **Area:**

$$Area = 4\pi R^2 \text{ Squ}$$

$$= 4\pi \left(\frac{\sqrt{8865}}{4}\right)^2 \text{ Squ}$$

$$Area = \frac{8865}{4} \pi \text{ squ.}$$

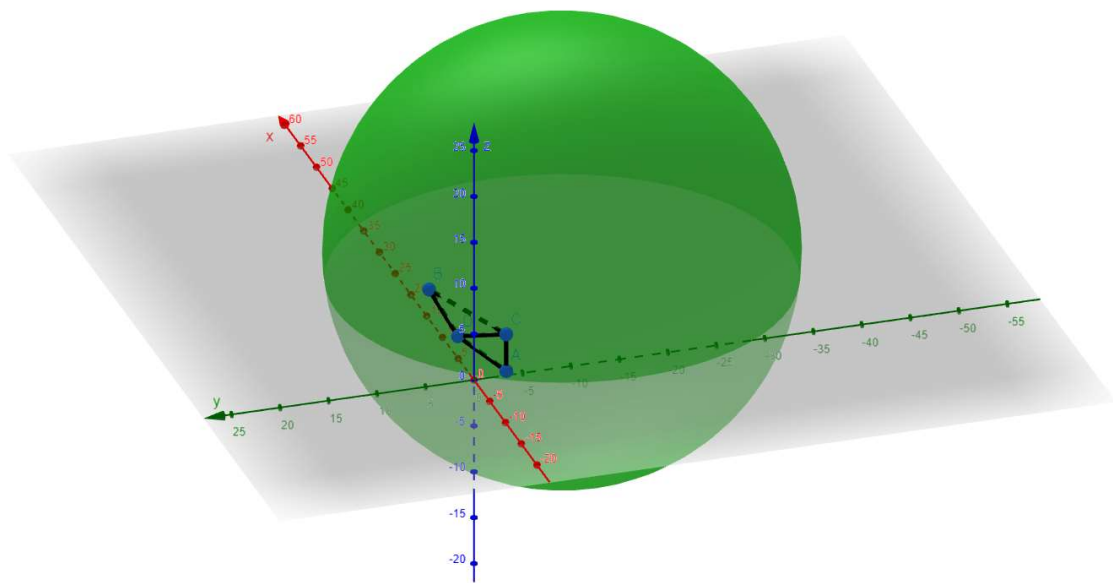
iii) **Volume:**

$$Volume = \frac{4}{3}\pi R^3 \text{ CU}$$

$$= \frac{4}{3}\pi \left(\frac{\sqrt{8865}}{4}\right)^3 \text{ CU}$$

$$Volume = \frac{2955}{16} \sqrt{8865} \pi \text{ CU}$$

H. Use 3D GeoGebra and plot the tetrahedron and its circumsphere.



Question 3:

1)

i. $y^2=4x$

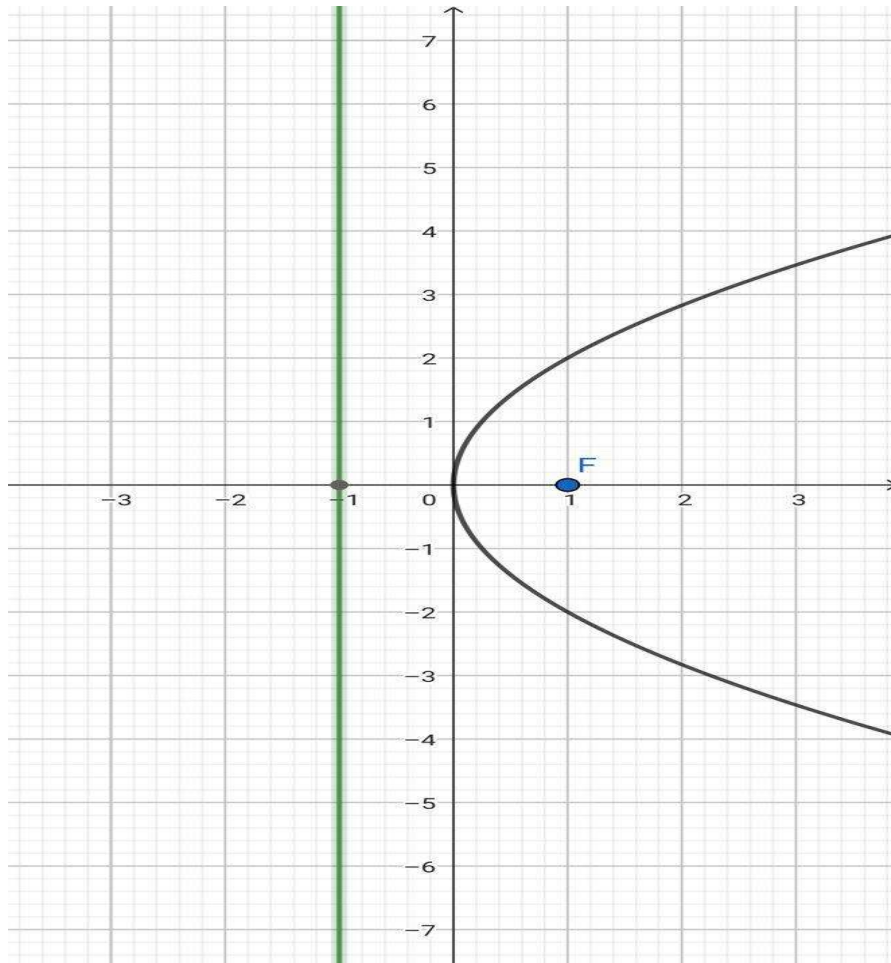
$$\begin{cases} x = \frac{t^2}{4} \\ y = t \end{cases} \rightarrow \text{parametric equation}$$

$2p=4$; $p=2 \rightarrow$ latus rectum

$$r = \frac{p}{1 \pm e \cos \theta} = \frac{2}{1 \pm \cos \theta} \quad e=1$$

Focus: $F(\frac{p}{2}, 0)$ $F(1,0)$

Directrix (d): $x = -\frac{p}{2} \rightarrow x = -1$



ii. $x^2=4y$

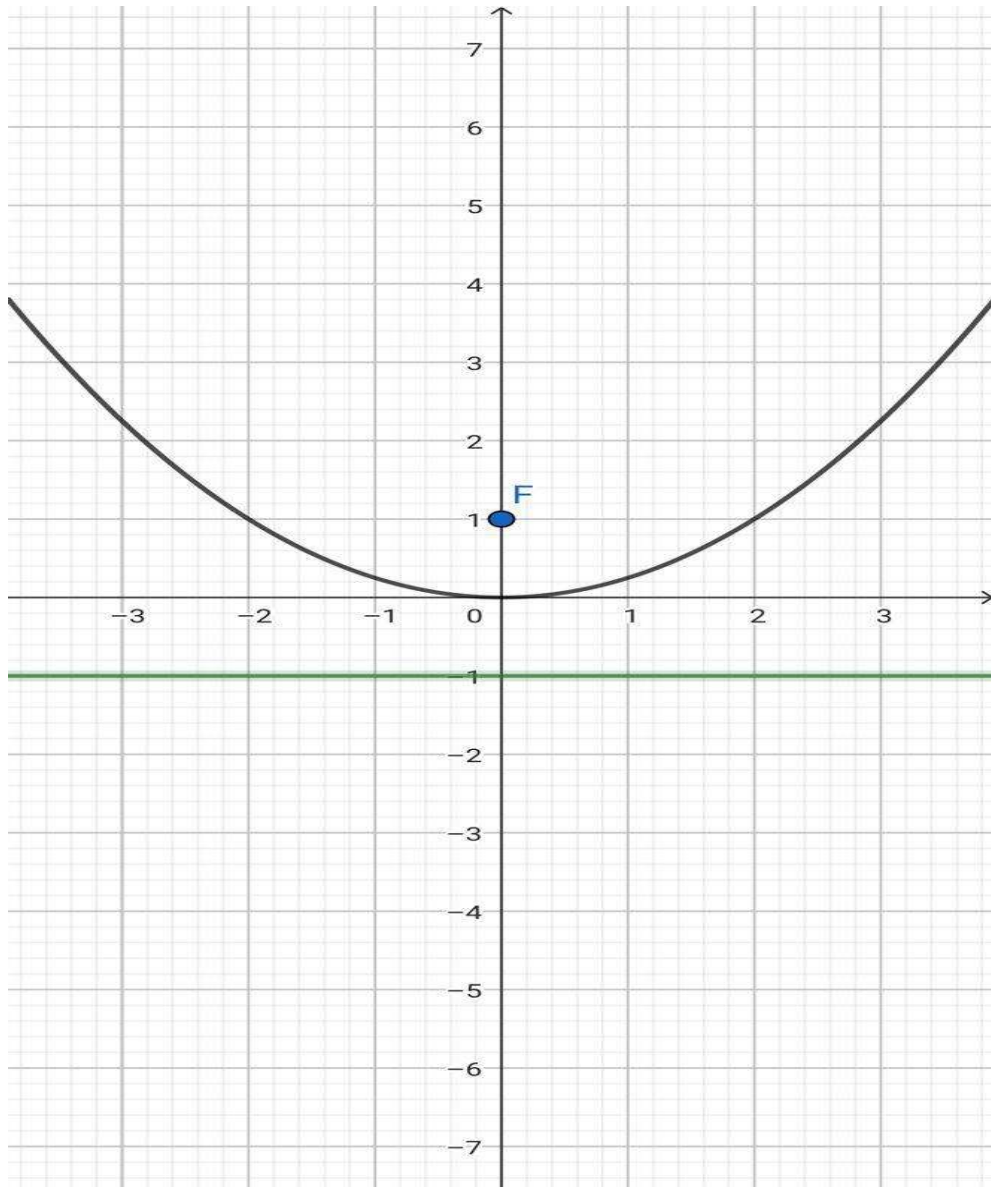
$$\begin{cases} x = \frac{t^2}{4} \\ y = t \end{cases} \rightarrow \text{parametric equation}$$

$2p=4; p=2 \rightarrow \text{latus rectum}$

$$r = \frac{p}{1 \pm e \cos \theta} = \frac{2}{1 \pm \cos \theta} \quad e=1$$

Focus: $F(0, \frac{p}{2})$ $F(0,1)$

Directrix (d): $y = -\frac{p}{2} \rightarrow y=-1$



2)

i. $\frac{x^2}{100} + \frac{y^2}{36} = 1$

$a=10$ $b=6$

$\begin{cases} x = 10 \cos t \\ y = 6 \sin t \end{cases} \rightarrow \text{parametric equation}$

$p = \frac{b^2}{a} = \frac{36}{10} \rightarrow \text{focal parameter}$

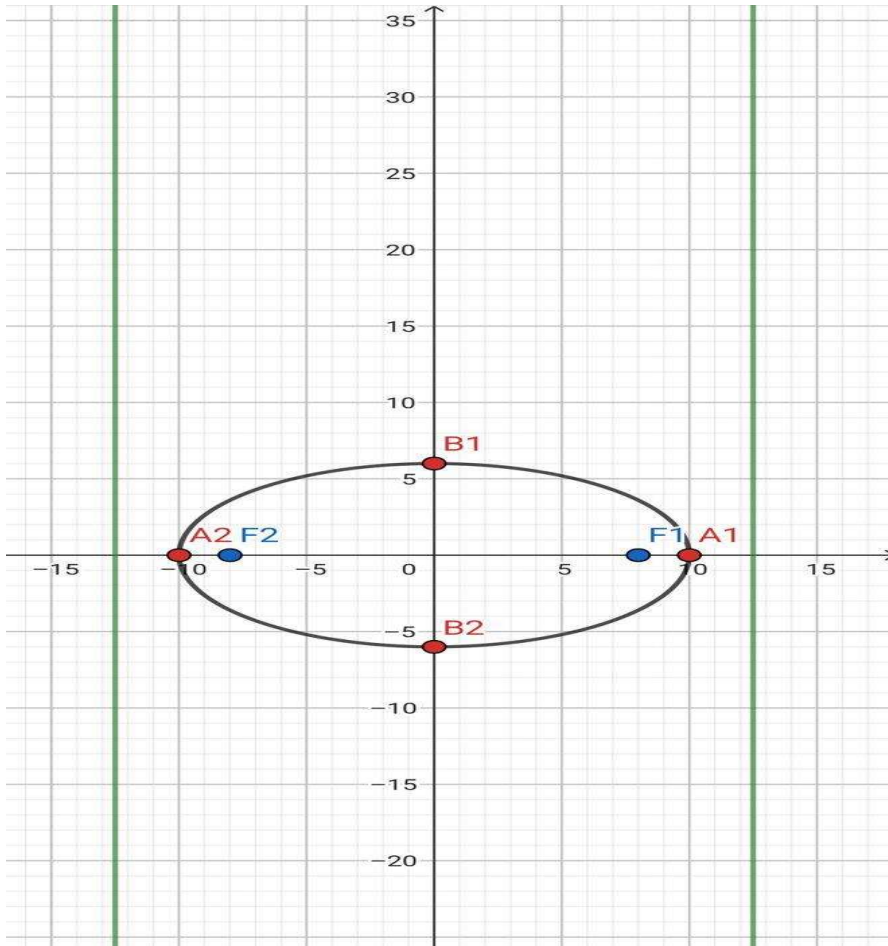
$e = \frac{c}{a}$ $c = \sqrt{b^2 - a^2} = \sqrt{100 - 36} = 8$ $e = \frac{8}{10}$

$$r = \frac{p}{1 \pm e \cos \theta} = \frac{36}{10(1 \pm \frac{8}{10} \cos \theta)} \rightarrow \text{polar equation}$$

Foci: $F(\pm c, 0)$; $F(\pm 8, 0)$

Vertices: $A_{1,2}(\pm a, 0)$; $A_{1,2}(\pm 10, 0)$ $B_{1,2}(0, \pm b)$; $B_{1,2}(0, \pm 6)$

Directrices ($d_{1,2}$): $x = \pm \frac{a^2}{c} = \pm \frac{100}{8}$



$$\text{ii. } \frac{x^2}{36} + \frac{y^2}{100} = 1$$

$$a=6 \quad b=10$$

$$\begin{cases} x = 6 \cos t \\ y = 10 \sin t \end{cases} \rightarrow \text{parametric equation}$$

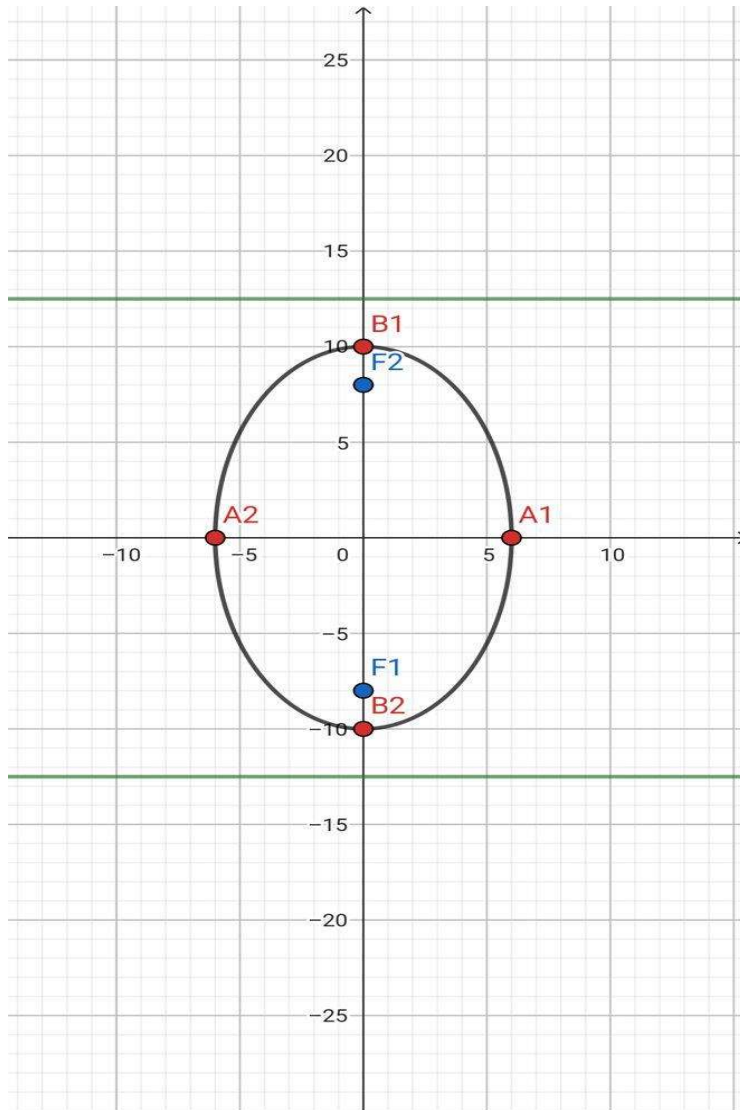
vertices: $A_{1,2}(\pm a, 0)$; $A_{1,2}(\pm 6, 0)$ $B_{1,2}(0, \pm b)$; $B_{1,2}(0, \pm 10)$

Directrices ($d_{1,2}$): $y = \pm \frac{a^2}{c} = \pm \frac{100}{8}$

$$C=8 \quad e = \frac{c}{b} = \frac{8}{10} \quad p = \frac{100}{6}$$

Foci: $F_{1,2}(0, \pm 8)$

$$r = \frac{100}{6(1 \pm \frac{8}{10} \cos \theta)} \rightarrow \text{polar equation}$$



3) i. $\frac{x^2}{9} - \frac{y^2}{16} = 1$

$a=3$ $b=4$

$$\begin{cases} x = 3 \sec t \\ y = 4 \tan t \end{cases} \rightarrow \text{parametric equation}$$

$$p = \frac{b^2}{a} = \frac{16}{3} \quad c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = 5 \quad e = \frac{c}{a} = \frac{5}{3}$$

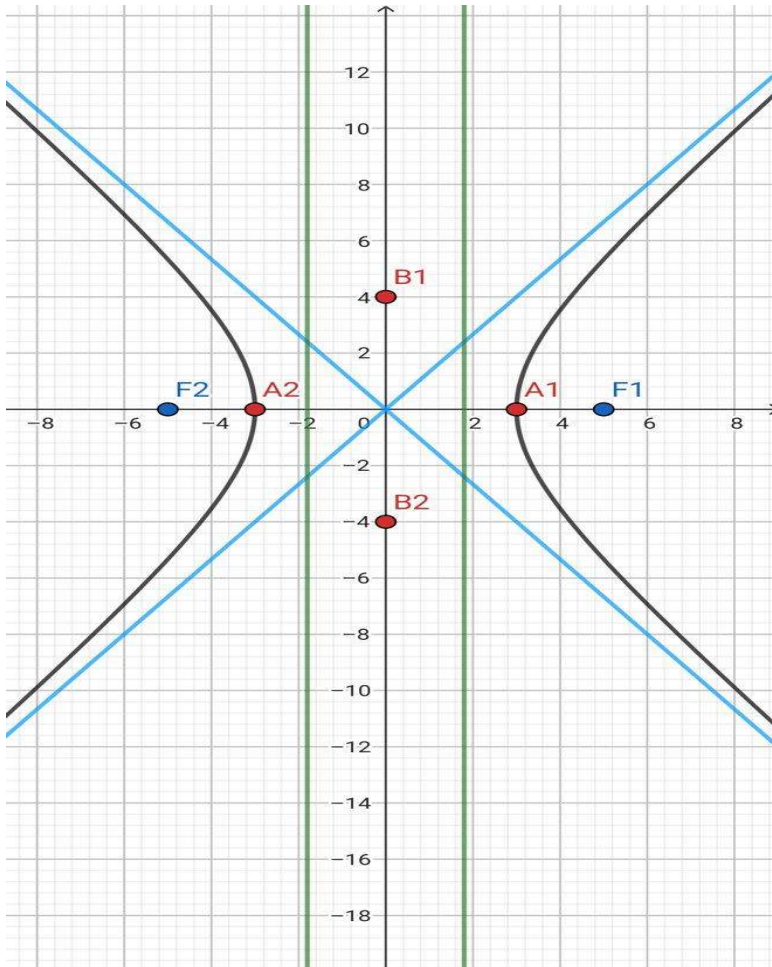
$$r = \frac{16}{3(1 \pm \frac{5}{3} \cos \theta)} \rightarrow \text{polar equation}$$

vertices: $A_{1,2}(\pm a, 0)$; $A_{1,2}(\pm 3, 0)$ $B_{1,2}(0, \pm b)$; $B_{1,2}(0, \pm 4)$

Foci: $F_{1,2}(\pm c, 0)$; $F_{1,2}(\pm 5, 0)$

Directrices ($d_{1,2}$): $x = \pm \frac{9}{5}$

Asymptote OA: $y = \pm \frac{4}{3}x$



4)

ii. $\frac{x^2}{9} - \frac{y^2}{16} = -1$

$a=3$ $b=4$

$\begin{cases} x = 3 \sec t \\ y = 4 \tan t \end{cases} \rightarrow \text{parametric equation}$

$p = \frac{b^2}{a} = \frac{16}{3}$ $c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = 5$ $e = \frac{c}{b} = \frac{5}{4}$

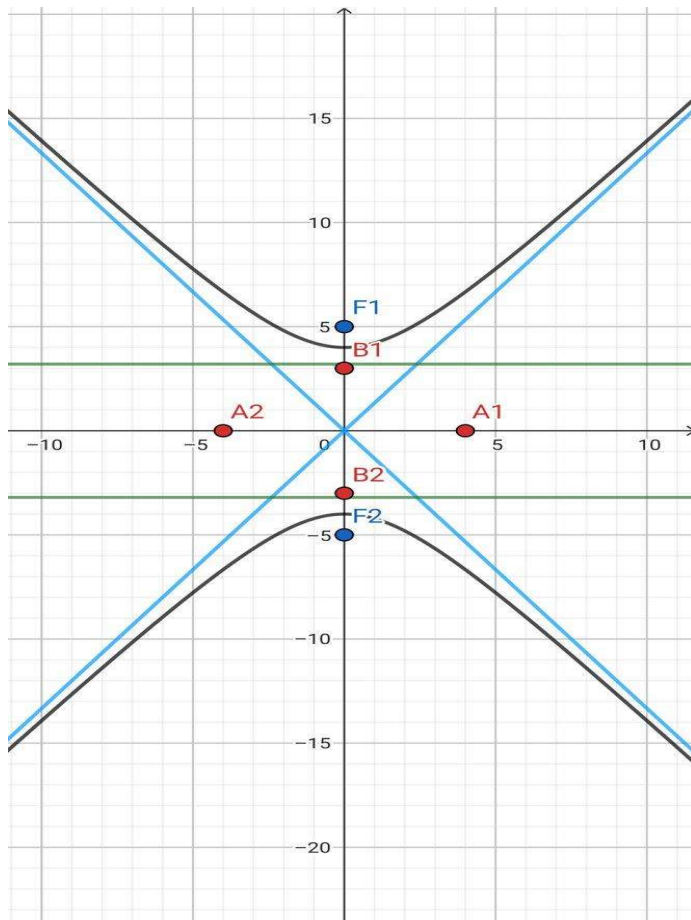
$r = \frac{16}{3(1 \pm \frac{5}{4} \cos \theta)}$ $\rightarrow$ polar equation

vertices: $A_{1,2}(\pm 4, 0)$ $B_{1,2}(0, \pm 3)$

Foci: $F_{1,2}(0, \pm 5)$

Directrices ($d_{1,2}$): $y = \pm \frac{16}{5}$

Asymptote OA: $y = \pm \frac{4}{3}x$



END OF OUR WORK